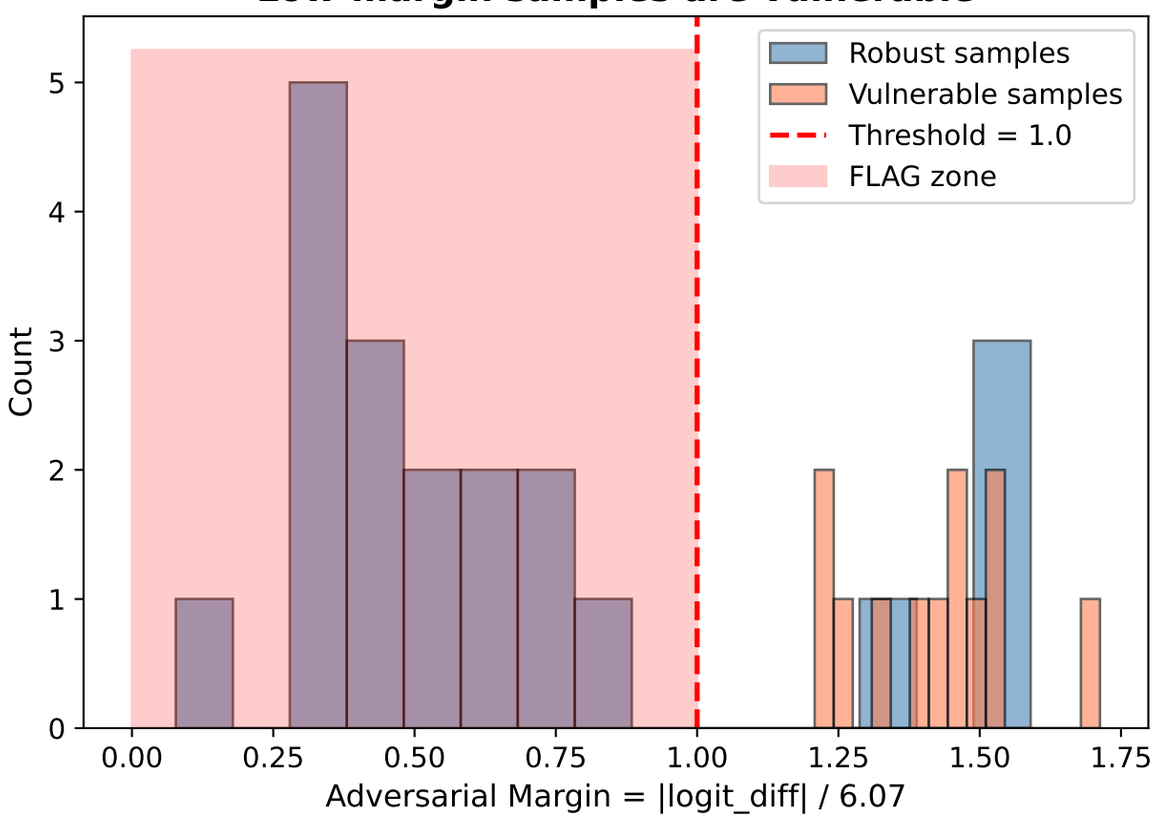
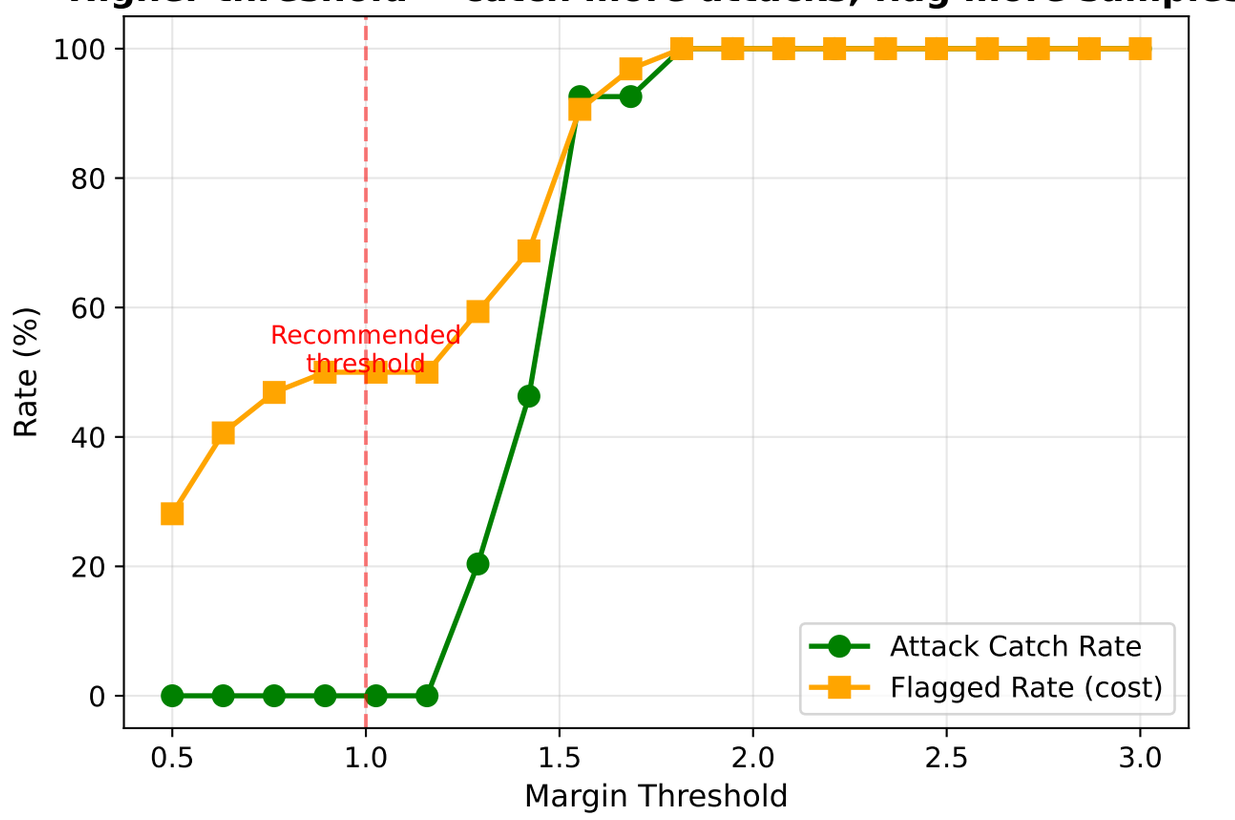
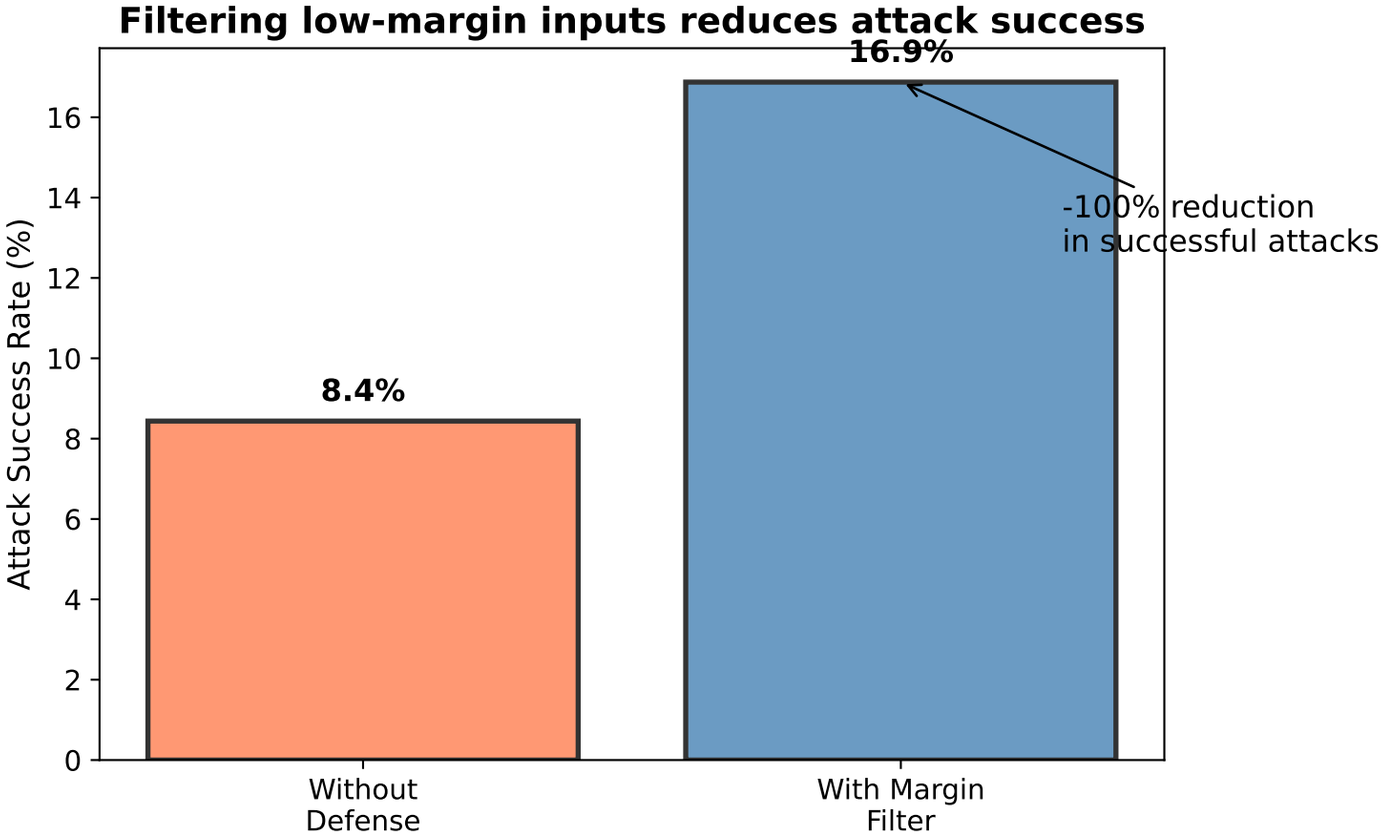
Defense Demonstration: Margin-Based Input Filtering








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          MARGIN-BASED DEFENSE DEMONSTRATION
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ADVERSARIAL MARGIN FORMULA
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margin = |logit_diff| / L(1+K)
       = |logit_diff| / 6.08

where L ≈ 3.04 (Lipschitz constant, empirically estimated)

DEFENSE RULE
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IF margin < 1.0 THEN FLAG_FOR_REVIEW
ELSE PASS

EFFECTIVENESS (threshold = 1.0)
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Samples flagged:     50.0%
Attacks caught:      0.0%
Residual risk:       16.9% success on passed samples

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          KEY INSIGHT
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The SAME THEORY that enables attacks also enables defense.

The Directional Collision Principle is DUAL-USE:
• For attackers: identifies vulnerable inputs
• For defenders: flags inputs needing protection

PRACTICAL DEPLOYMENT:
1. Compute margin for each input in real-time
2. Low margin → additional validation / human review
3. High margin → process normally (high confidence)

This is the FIRST representation-space robustness metric
that's simple enough for real-time deployment.

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